SEARCH & RECOVERY (SAR) FORENSIC GEOPHYSICS

GPR Equipment Calibration & Field Survey Log Sheet

Target Site: Target Delta / LT-HAZARD-03	**Date/Time:** ____/____/2026 ____:____	**Operator ID:** _____
GPR System: 400 MHz / 900 MHz PulseEKKO	**Grid ID:** 10m x 10m Transect ____	**Weather/Soil:** Saturated Silt / Clay

1. TWO-WAY TRAVEL TIME (TWT) & VELOCITY CALIBRATION (HYPERBOLA FITTING)

Step / Parameter	Target Metric	400 MHz Setting	900 MHz Setting	Field Verified Value
Test Stake Hyperbola Fitting	TWT (ns) / Velocity (m/ns)	v ~ 0.080 m/ns	v ~ 0.080 m/ns	[ ] _____ m/ns
Relative Dielectric Permittivity (Er)	Saturated Silt Baseline	Er = 12.0 - 16.0	Er = 12.0 - 16.0	Er = [ ] _____
Time Zero (T0) Offset	Antenna Air-Ground Contact	0.0 ns Offset	0.0 ns Offset	Calibrated: [ ] YES
Total Time Window / Depth Range	Depth Penetration	75 ns (~3.0m Depth)	30 ns (~1.2m Depth)	[ ] _____ ns
Trace Sampling / Step Interval	Spatial Resolution	0.05 m Step	0.02 m Step	[ ] _____ m

2. TRANSECT ANOMALY RECORDING LOG (400 MHz & 900 MHz DUAL SCAN)

Transect #	Pos (m)	Depth (m)	Hyperbola Polarity	Ringdown?	Circularity	Target Match Category	Action Required
LINE-01	3.2m	0.85m	Inverted (Phase Flip)	No	0.88	Human Cranial Vault (Skull)	Mark for HRD K9 Probe
LINE-02	4.5m	1.25m	Normal High Amplitude	Yes (Metallic)	N/A	Footwear / Metal Hardware	Metal Detector Cross-Check
LINE-03	____m	____m	Normal / Inverted	Yes / No	_____	_____	_____
LINE-04	____m	____m	Normal / Inverted	Yes / No	_____	_____	_____
LINE-05	____m	____m	Normal / Inverted	Yes / No	_____	_____	_____
LINE-06	____m	____m	Normal / Inverted	Yes / No	_____	_____	_____

3. HRD K9 VAPOR PROBING & FIELD VERIFICATION PROTOCOL

[ ] Soil Dielectric Check: Verify velocity profile er = 12-16 using ground probe before calculating target depth.

[ ] Hollow Vault Resonance: Confirm inverted phase hyperbola at 0.85m depth on Sliceview horizontal slice map.

[ ] HRD K9 Soil-Gas Probing: Insert 0.5m gas sampling pin upwind of anomaly prior to surface disruption.

[ ] Metallic Hardware Check: Confirm vertical ringdown signature with multi-frequency ground magnetometer.

[ ] Excavation Handoff: Pin physical stake at exact GPS coordinate; notify Lead Forensic Anthropologist.

GPR Operator Signature: _____ **HRD K9 Handler Signature:** _____